

Solubility Curve Lab

Teacher

- Turn on room fan
- Remind them not to reach OVER a flame
- Set a timer to press Stops & turn off the fan

Alter lab in following way (groups of 3-4)

- Use a gentle BLUE flame (orange will result in black t.t.)
- Need an empty larger Beaker to have a spot to put t.t. down or large t.t. Holder
- Be MOVING flame slowly underneath for even heating
- Angle away from classmates
- Only heat until all is dissolved (NOT 90C) the first time
- Turn of the Bunsen burner at this point
- Use a coloured paper behind t.t. to watch for crystals
- If you are fast at adding water, it will all re-dissolve WITHOUT further heating.
- **TASKS TO DIVIDE:** holding test tube, watching for crystals, recording temperature, adding more water

Final few data points:

You can hold the test tube intermittently to get the temperature to lower faster.

If time allows you could also place the tube in cold or **ice water**

IN LAB - GROUPS OF 4

- PRACTICE with bulbs and pipettes first if needed
- Draw up 3 or more ml of water, remove tip from water and remove bulb, covering end with finger and releasing until exactly 2 ml. Do this IN ADVANCE so can quickly add it to test tube
- TASKS: holding test tube, watching for crystals, recording temperature, adding more water
- Need TWO beakers -> one of distilled water to ADD and empty one to hold test tube as needed.

Clean up:

- Dispose of liquid in provided beaker
- Wash test tubes with a scrub brush in sink with soapy water
- Wash all other glassware and return to where you got it, ensure all beakers are cleaned etc.

Equipment list

Ammonium chloride small beakers with scoopulas

Retort stands

Thermometers

Tt clamps (and pinchers)

Hot mits

10 ml pipettes & bulb, 10 ml cylinders & droppers, FYI Plastic pipette bulbs have a 2 ml marking so can also be used.

Large & wide tt

Enough beakers for distilled water and a holder

Test tube holders